The ten minimum universal sets of MacMahon cubes

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Abstract

We classify universal collections of twelve six-coloured MacMahon cubes, without requiring internal face matching. A row-sum bound in Conway's tableau forces a symmetric matrix with every row and column sum equal to two. Corner coverage then forces two disjoint triangles. This proves the ten-set uniqueness conjecture stated by Johnson and Roldán, and also excludes additional minimum collections containing repeated cube varieties. The argument is elementary; exhaustive computation is used only as a check and to obtain additional target-count distributions. The minimum-size theorem, the complete classification and their discrete geometric interpretation are formalised in Lean 4.

Status and transparency. This research manuscript and its Lean formalisation were developed with AI assistance under the author's direction. The formal proofs have been checked by Lean; independent human expert review and publication priority beyond the cited sources remain outstanding. Details of the proof interface and AI involvement appear below.

1 Conventions and tableau facts

There are thirty rotational varieties of cubes whose six faces have six different colours. A collection is *universal* if eight of its physical cubes can build a $2 \times 2 \times 2$ model of every target variety. Interior face colours need not match. Repeated varieties are allowed in the theorem; the conjecture in [1] concerns sets of distinct varieties.

Label Conway's tableau by ordered pairs ij of distinct elements of $L = \{A, B, C, D, E, F\}$; ij corresponds to the paper's upper/lower-case name such as Ab. Each cube has eight oriented corner types. A corner type is an ordered triple of face colours modulo cyclic permutation, with a fixed handedness. There are forty such types.

We use the following finite description of the standard tableau. A corner type corresponds to a partition of L into two triples, with a directed 3-cycle on each triple. Its six incident cube varieties are precisely the six directed edges of these cycles. All ten partitions and all four choices of cycle directions occur. This can be verified from Table 3 of [1]; Table 1 below groups that table by the underlying partition. The cube ij has exactly the eight corner types containing the arc $i \to j$. This also gives a compact, purely finite model for the corner incidence.

Two distinct varieties ij and uv share no corner exactly when $i = u$, $j = v$, or $(u, v) = (j, i)$. In every other case they share two corners. Indeed, two distinct arcs in the same directed 3-cycle must form a directed path; this fixes that cycle, leaving two orientations of the other one. For two disjoint arcs, the remaining two labels can be assigned to their respective triples in two ways, each determining both cycle directions. Common tails, common heads, and reversed arcs cannot coexist in directed 3-cycles. The target itself shares all eight corners.

Consequently each cube is usable for exactly 21 targets. Also, assigning eight distinct physical cubes to the eight target corner types is both necessary and sufficient: matching an oriented corner permits a proper rotation matching the three exposed faces. There are no internal constraints.

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Partition	Four oriented corner types
$ABC \mid DEF$	256, 143, 134, 265
$ABD \mid CEF$	465, 132, 123, 456
$ABE \mid CDF$	345, 126, 162, 354
$ABF \mid CDE$	235, 164, 146, 253
$ACD \mid BEF$	364, 125, 152, 346
$ACE \mid BDF$	356, 142, 124, 365
$ACF \mid BDE$	153, 246, 264, 135
$ADE \mid BCF$	165, 243, 234, 156
$ADF \mid BCE$	136, 254, 245, 163
$AEF \mid BCD$	263, 145, 154, 236

Table 1: The corner types associated with each pair of triples. Directions are recovered from the six cube names containing the corner in the standard tableau. Corner numbers use the colour labels of [1].

2 Necessity

Let a_{ij} be the number of copies of variety ij in a collection of size K, with $a_{ii} = 0$. Write $r_i = \sum_j a_{ij}$ and $c_j = \sum_i a_{ij}$.

Lemma 1 (Row bound)**.** *For a universal collection,*

$$r_i + c_j - 2a_{ij} + a_{ji} \le K - 8 \qquad (i \ne j), \qquad 3r_i \le 4K - 40, \quad 3c_i \le 4K - 40.$$

Proof. For target ij, the unusable cubes are those in its row apart from ij, those in its column apart from ij, and the copies of ji. These classes are disjoint and total $r_i + c_j - 2a_{ij} + a_{ji}$. At least eight cubes must remain usable, proving the first inequality. Summing over the five $j \ne i$ gives

$$5r_i + (K - c_i) - 2r_i + c_i = 3r_i + K \le 5(K - 8).$$

The column bound follows by the analogous sum. □

Lemma 2. *A universal collection of size twelve consists of one copy of every directed edge within each of two disjoint triples of tableau labels.*

Proof. The row bound gives $r_i \le \lfloor 8/3 \rfloor = 2$. As the six row sums total twelve, all equal two. The same holds for the columns. If $a_{ij} = 0$, the first inequality now gives $4 + a_{ji} \le 4$, so $a_{ji} = 0$. Thus support is symmetric.

Consider the undirected support graph on L. Every vertex has degree one or two. If a vertex i has degree one, with neighbour j, then $a_{ij} = 2$; column i and support symmetry give $a_{ji} = 2$, so this is an isolated double-copy pair. Every degree-two component is a simple cycle, with one copy in each direction on every edge.

On six vertices the only possibilities are two triangles, a 6-cycle, a 4-cycle and a double-copy pair, or three double-copy pairs. All except two triangles are bipartite with parts of size three. Choose these parts as the two triples defining a corner type. Every directed edge of that corner lies within a part, whereas every selected variety joins the parts. Thus the collection contains no copy of that corner type. A target having that corner cannot be built, contradicting universality. Only two triangles remain, and their arc multiplicities are all one. □

3 Sufficiency and classification

Lemma 3. *For any partition $L = P \sqcup Q$ into triples, the twelve directed edges within P and Q form a universal set.*

Proof. Fix target ij. Its corners are indexed by a third label $k \notin \{i, j\}$ and either direction of the cycle on the complementary triple. The first cycle is $i \to j \to k \to i$.

Suppose $i, j \in P$, and write $P = \{i, j, p\}$. For each $k \in Q$, the complementary cycle contains one arc between the two labels of $Q \setminus \{k\}$. Assign that cube to the corner. As k and the cycle direction vary, these are the six distinct directed edges within Q. The remaining two corners have $k = p$; fill them with jp and pi, both of which belong to both corner types. This gives eight distinct cubes.

Now suppose $i \in P$, $j \in Q$, with $P \setminus \{i\} = \{u, v\}$ and $Q \setminus \{j\} = \{x, y\}$. For the four corners with $k \in \{u, v\}$, the usable selected varieties are ki and one of xy, yx. Their incidence is that of the four edges of a $K_{2,2}$ with its four vertices. Orient this 4-cycle cyclically to assign a different incident variety to each corner. For the other four corners, the analogous four varieties are jx, jy, uv, vu. The two groups of four varieties are disjoint, completing the construction. □

Theorem 4. *The minimum size of a universal collection is twelve. Exactly ten labelled collections attain that size, even when repeated varieties are permitted. They are the two-triangle sets described above. In particular, Conjectures 4 and 8 of Johnson and Roldán hold.*

Proof. Each physical cube is usable for 21 targets, so universality requires $21K \geq 30 \cdot 8 = 240$, hence $K \geq 12$. This lower bound is due to Haraguchi [2]. The preceding lemmas give necessity and sufficiency at twelve. There are $\frac{1}{2}\binom{6}{3} = 10$ unordered partitions into triples. □

4 Computational checks and additional results

An exhaustive enumeration of all $\binom{30}{12} = 86,493,225$ subsets returns exactly the ten sets above. The implementation also reproduces the published distribution for all $\binom{30}{8} = 5,852,925$ eight-cube sets. Two different feasibility algorithms are used: bipartite matching, and a graph-component test. In the latter, every usable non-target cube is an edge joining its two compatible target corners. A connected tree component has deficit one; a component containing a cycle has deficit zero. Thus a target is buildable precisely when the number of tree components does not exceed the number of available target copies (zero or one in the subset enumerations). The table-based implementation precomputes all 2^{20} edge subsets for each target. No symmetry reduction is used in the exhaustive subset loops.

Independent geometric code generates all 24 proper cube rotations, all 30 varieties and all 40 corner types from face-colour permutations. It checks the tableau, the directed-triangle correspondence, and explicit physical placements for all 300 set–target pairs. Independently enumerating the 70 simple 2-regular graphs gives 60 six-cycles, each building 18 targets, and the ten universal triangle-pair sets. The ten sets form one colour-permutation orbit, with stabilizer order 72.

For sets of distinct varieties the exhaustive target-count ranges are:

set size	8	9	10	11	12	13
minimum targets	0	0	1	2	8	10
maximum targets	5	9	20	22	30	30

No set of size at most twelve builds exactly 29 targets. At size thirteen, 827,250 sets build exactly 29 targets and 53,100 are universal. These enumeration results address further questions in Section 6 of [1]; unlike the uniqueness theorem, their upper bounds rely on the complete computational enumeration. Full histograms, example masks, source code and independent checkers accompany this draft.

5 Formal verification and reproducibility

The Lean 4 development accompanies this manuscript at https://github.com/jasondavies/macmahon-min-universal-sets. It is pinned to Lean 4.35.0 and mathlib v4.35.0. The main results are `MacMahon.minimum_size`, `MacMahon.physical_exactly_ten`, and `MacMahon.physical_iff_two_triangles`.

An inventory is a natural-number multiplicity for each ordered pair of six tableau labels, with zero diagonal. A construction is an injective assignment of eight physical copies to the eight positions, each with a proper rotation matching its three exposed faces. The formalisation proves that the rotation table is exactly the set of determinant-positive signed-axis permutations, and that the thirty representatives and their rotations give all 720 assignments of six distinct colours to six faces, without duplication. It also proves the equivalence of corner matching and exposed-face matching.

The inequalities and symmetry argument are proved for arbitrary natural multiplicities. The remaining classification is checked through 130 arithmetic proof branches, split across fifteen modules. All finite checks use kernel reduction, rather than a trusted external computation. Explicit witnesses certify all 300 minimum-set–target constructions. The proof library contains no added axioms or unfinished proofs; guarded audits allow only Lean's standard propositional extensionality, classical choice and quotient axioms. The larger target-count enumerations above are not included in these formal theorems.

For Palomar verification, `Challenge.lean` imports only mathlib and states ten advertised results with intentional proof holes, independently of the proof library. `Solution.lean` imports the completed proofs. The comparator configuration checks equality of the statements, their permitted axioms, and kernel acceptance. Local verification outputs and reproduction commands are recorded in the repository; a successful local check is distinct from registry acceptance.

AI disclosure. OpenAI Codex assisted with the investigation, compiled solvers, geometric checks, proof argument, Lean development, manuscript and submission packaging, under the author's direction. No claim of independent human expert review is made here. The proof generators are not trusted components: their output must be accepted by Lean's kernel.

References

[1] I. Johnson and E. Roldán, *Solution Numbers for Eight Blocks to Madness Puzzle*, arXiv:2407.13208v1 (2024). https://arxiv.org/abs/2407.13208.

[2] K. Haraguchi, *On a generalization of "Eight Blocks to Madness" puzzle*, Discrete Mathematics **339** (2016), 1400–1409. Preprint arXiv:1408.3696v2. https://arxiv.org/abs/1408.3696.